

Ahmed Thabet

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Building and scaling the world's largest decentralized P2P cloud infrastructure for over 8 years.

Experience

- *Codescalers*: **Engineering Manager** (2023 – Present)
Leading multiple engineering teams responsible for the design, development, and operation of the world's largest decentralized P2P cloud infrastructure.
Ownership includes technical strategy, architectural decisions, roadmap planning, delivery, hiring, mentoring, and cross-team coordination across the Threefold Grid ecosystem.
- *Codescalers*: **Software Development Team Lead** (2020 – 2023)
Led the development of the largest P2P cloud infrastructure in the world, including the Jumpscale cloud framework, the 3Bot virtual system administrator, and the Virtual Data Center (VDC).
Focused on system architecture, hands-on development, code quality, and delivery.
- *Codescalers*: **Software Engineer** (April 2016 – 2020)
Designed and implemented cloud tooling for infrastructure management, including virtual machines, containers, and automation workflows.
- *Otlob.com*: **Data Analyst** (Freelance)
Extracted audio fingerprints from large-scale video datasets to evaluate marketing campaign performance.
- *Orascom Telecom*: **Web Developer** (Freelance)
Built a PHP/CodeIgniter application to manage internal server air-conditioning data, replacing Excel-based workflows.

Work

Codescalers (Engineering Manager)

I lead the development of the largest P2P cloud and the ecosystem of tooling around it.

- *ZOS*: Leading the development of Zero-OS, the operating system powering the Threefold Grid (Go)
- *AIBroker*: Leading the development of an AI Broker built on top of multiple LLM providers and MCP servers (Rust, Go)
- *KubeCloud*: Highly available Kubernetes platform running on the Threefold Grid (Go, Vue.js, PostgreSQL, Docker)

- *EWF*: Lightweight, embeddable workflow framework for Go applications, enabling resilient, stateful, multi-step processes (Go)
- *ContentHive*: Content-addressed storage system inspired by IPFS (Go)
- *JS-NG*: Led the development of the cloud automation and configuration framework (Python, Gevent, Docker)
- *JS-SDK*: Led the development of the Threefold Grid software toolkit and the Threebot application server (Python, Gevent, Docker)
- *VDC*: Led the development of the Virtual Data Center platform (Kubernetes clusters, S3, Prometheus/Grafana monitoring, management dashboards) on the Threefold Grid (Python, Gevent, Stellar blockchain, Kubernetes, Ansible, Grafana)
- *VDC Marketplace*: Led the development of the Virtual Data Center Marketplace, enabling one-click application deployments
- *Threefold Marketplace*: Led the development of the Threefold Marketplace, enabling one-click application deployments on the grid
- *Terraform/OpenTofu Plugin*: Led the development of the Terraform provider for the Threefold Grid (Go)
- *Golang SDK*: Led the development of the Go SDK, enabling delivery of multiple grid-based projects
- *TFRobot*: Led the development of a mass-deployment tool for large VM fleets on the Threefold Grid
- *Gridify*: Led the development of a Heroku-like deployment tool for projects on the Threefold Grid
- *Farmerbot*: Led the development of a power-management and operations tool for grid farmers
- *Typescript SDK*: Led the development of the TypeScript SDK to enable web developers to integrate with the Threefold Grid
- *Weblets*: Svelte-based reusable web components for rapid application development on the Threefold Grid
- *Go farmer*: Cross-platform GUI application for farmers to manage their farms (Go, Fyne)
- *publishingtools*: Led the development of a web publishing framework for wikis, blogs, and websites, primarily for Threefold (Crystal, Kemal, Svelte)
- *CRM*: Led the development of a generic CRM system for the Threefold Token Foundation (Python, Flask, SQLAlchemy, GraphQL, PostgreSQL, Docker)
- *0-Disk*: Developed a Network Block Device (NBD) server with multiple storage backends; later replaced by 0-db (Tarantool, Lua, fio)
- *AYS*: Agentless Infrastructure-as-Code orchestration system built on Jumpscale; implemented event systems, background tasks, and service templates (OwnCloud, S3, PHP, Nginx)
- *OpenVCloud*: Cloud platform based on libvirt and Python; developed monitoring, alerting, and hardware health-check tooling
- Additional work: OAuth provider for *Gogs*, DMIDecode parser, rtinfo integration, internal development workflow automation, GitHub-Gogs synchronization, and documentation generators for Jumpscale 7 and 8

Open Source Projects

- pygundb: GUN.js database server and client (Python)
- tcprouter: TCP router with SNI-based routing (Go)
- tfgateway: Threefold gateway built on tcprouter and CoreDNS Redis plugin
- nim-servy: Web framework for Nim
- vscode-capnp: Cap'n Proto extension for VS Code
- coredns-redis: Redis plugin for CoreDNS
- flask_itsyouonline: OAuth2 middleware for Flask
- python-pcre: PCRE bindings for Python using ctypes
- raqeeb: PID 1 replacement and advanced alternative to Foreman
- pyparsec: Parser combinator library for Python
- helloED: Pluggable, fast PyGTK text editor
- stellargui: Lightweight Stellar blockchain wallet
- Pida IDE: Contributed RegexToolkit and Pastebin plugins
- python-sonic-client: Python client for Sonic search backend
- nim-assets: Bundle assets into a single Nim binary
- nim-redisclient: Redis client and parser for Nim
- nim-parsec: Parser combinator library for Nim
- raml2spore: Generate SPORE clients from RAML specifications

Publications / Translations

- Nim Days: Practical applications and libraries in Nim
- Python: Extensive Guide
- Introduction to Ruby
- Translations: PyQt4, PyGTK, Mono IronPython, and Cairo tutorials translated into Arabic

Skills & Technologies

- **Languages:** Python, Go, Ruby, Rust, Java, PHP, Nim, TypeScript, Crystal, C, Bash, Scheme, Haskell, Scala, OCaml
- **Databases:** MySQL, PostgreSQL, MongoDB, Redis, InfluxDB
- **APIs:** REST, Swagger, RAML, GraphQL, gRPC, Cap'n Proto
- **Web Frameworks:** Flask, Flask-Admin, Sanic, Django, FastAPI, Bottle, CodeIgniter, Laravel, CakePHP, Yii, Rails, Sinatra, Gin, Gorilla, Echo
- **Frontend:** Svelte, Sapper, Vue, React
- **Infrastructure:** Linux, Vagrant, Docker, Docker Compose, Docker Swarm, Kubernetes, K3s, Helm, Terraform, Ansible, Prometheus, Grafana, CoreDNS, Nginx, Caddy, Traefik

Education & Languages

- BSc in Electrical Engineering, Helwan University, Cairo, Egypt

- English: Business proficiency (speaking, reading, writing)
- German: Limited working proficiency (B1.1)